## Introduction

Data mining is the process of discovering useful patterns, relationships and knowledge from large amounts of data. It helps organizations analyse information and make better decisions.

Data mining is widely used in industries such as banking, healthcare, telecommunications, education and e-commerce.

The main purpose of data mining is to extract meaningful information from raw data using different algorithms and techniques.

Data preprocessing is an important stage in data mining. It includes data cleaning, integration, transformation, reduction and discretization. These techniques improve data quality and help produce accurate results.

## Objectives

1. To understand different data mining functionalities.
2. To study the Knowledge Discovery in Databases (KDD) process.
3. To calculate and analyse data dispersion measures.
4. To identify and handle missing, noisy and inconsistent data.
5. To perform data integration and normalization.
6. To understand data reduction and discretization techniques.
7. To design and evaluate an improved data preprocessing pipeline.

## Q1 – Matching Business Problems to Mining Functionalities (5 Marks)

### 1.1 Introduction

Data mining functionalities are used to discover different types of patterns from large datasets.

TeleNova is a telecommunications company with 8 million subscribers. It wants to analyse customer behaviour, identify usage patterns and predict customer churn.

Different data mining techniques are required to solve these business problems.

### 1.2 Mapping Business Problems to Data Mining Functionalities

| Problem | Data Mining Functionality | Learning Type | Knowledge Produced |
|---|---|---|---|

| | | | |
|---|---|---|---|
| P1: Find plans and add-ons purchased together | Association | Unsupervised | Relationships and frequent itemsets |
| P2: Predict whether customers will leave | Classification / Prediction | Supervised | Customer churn prediction |
| P3: Divide subscribers into usage segments | Clustering | Unsupervised | Natural customer groups |
| P4: Identify unusual SIM behaviour | Outlier Analysis | Unsupervised | Abnormal usage patterns |
| P5: Compare high-value and low-value customers | Class / Concept Description | Supervised or descriptive | Customer characteristics |
| P6: Study monthly data usage over 24 months | Evolution Analysis | Time-based analysis | Trends and changes over time |

### 1.3 Explanation of Each Problem and Algorithms

**P1 – Association Analysis**

Association analysis identifies relationships between products or services that are frequently purchased together.

Example: Customers who purchase an unlimited data plan may also purchase an international calling add-on.

Algorithms:

1. Apriori Algorithm – Identifies frequent itemsets and association rules.
2. FP-Growth Algorithm – Finds frequent patterns efficiently using an FP-tree.

Required attributes:

- Customer ID
- Plan ID
- Add-on ID
- Purchase date
- Transaction ID

Data source: Telecom subscriber transaction database or data warehouse.

**P2 – Customer Churn Prediction**

Classification is used to predict whether a customer will leave the telecom network.

The model learns from historical customer records labelled as churn or non-churn.

Algorithms:

1. Decision Tree – Classifies customers using decision rules.
2. Random Forest – Combines multiple decision trees to improve prediction.

Required attributes:

- Customer ID
- Monthly bill
- Call duration
- Data usage
- Complaint history
- Recharge frequency
- Contract duration
- Previous churn status

Data source: Customer database and historical telecom warehouse.

**P3 – Customer Segmentation**

Clustering divides subscribers into groups based on similar usage behaviour without predefined labels.

Example:

- High-data users.
- Low-data users.
- Frequent callers.
- Occasional users.

Algorithms:

1. K-Means Clustering – Groups customers according to similarity.
2. DBSCAN – Identifies dense groups and customers that do not belong to any major group.

Required attributes:

- Monthly data usage
- Call duration
- Recharge amount

- SMS count
- Monthly expenditure

Data source: Telecom usage database.

**P4 – Outlier Analysis**

Outlier analysis identifies records that are significantly different from normal behaviour.

Example: A SIM card suddenly makes hundreds of international calls within a short period.

Algorithms:

1. Isolation Forest – Identifies unusual observations.
2. LOF (Local Outlier Factor) – Detects records that are less dense than their neighbouring records.

Required attributes:

- SIM ID
- Call frequency
- Call duration
- International call count
- Location
- Data usage

Data source: Telecom network usage logs.

**P5 – Class and Concept Description**

Concept description summarizes the characteristics of different customer groups.

Example: High-value customers may have higher monthly bills, greater data usage and longer subscription periods than low-value customers.

Algorithms:

1. Decision Tree – Identifies important attributes that distinguish customer groups.
2. OLAP / Attribute-oriented induction – Summarizes customer information at different levels of detail.

Required attributes:

- Customer segment
- Monthly revenue
- Data usage

- Call duration
- Subscription period
- Recharge amount

Data source: Customer data warehouse.

**P6 – Evolution Analysis**

Evolution analysis studies how data changes over time.

Example: Monthly data usage may increase during holidays or decrease during certain months.

Algorithms:

1. Time-Series Analysis – Identifies trends and seasonal patterns.
2. Linear Regression – Estimates changes in usage over time.

Required attributes:

- Customer ID
- Month
- Data usage
- Call duration
- Recharge amount

Data source: Historical telecom data warehouse.

### 1.4 Why a Single Algorithm Cannot Solve All Problems

The marketing manager proposes purchasing one all-purpose mining algorithm for all six problems.

This is not a suitable decision because every problem has a different objective and requires a different mining functionality.

For example:

- Association algorithms identify relationships between purchased services.
- Classification algorithms predict customer churn.
- Clustering algorithms group customers based on similarity.
- Outlier algorithms identify unusual SIM behaviour.
- Evolution analysis identifies trends over time.

A clustering algorithm cannot directly replace an association algorithm, and a classification algorithm requires labelled training data.

Therefore, TeleNova should select different algorithms according to the business problem.

### 1.5 Conclusion

Different data mining functionalities help TeleNova understand customer behaviour, identify unusual activities and predict future trends. Using appropriate algorithms for each problem provides more accurate and useful results.

## Q2 – The KDD Process, System Architecture and Mining Issues (5 Marks)

### 2.1 Introduction

MediCore is a hospital network that wants to identify the factors responsible for patients being readmitted within 30 days.

Patient information is stored in different systems, including admission, laboratory, pharmacy and billing databases.

The KDD process helps convert this raw data into useful knowledge.

KDD stands for Knowledge Discovery in Databases.

### 2.2 Steps in the KDD Process

The KDD process contains five major steps.

Figure 2.1: KDD Process

**1. Data Selection**

Data selection identifies the information required for analysis.

For MediCore, data is collected from:

- Patient admission records.
- Laboratory reports.
- Pharmacy records.
- Billing information.
- Previous hospital visits.

**2. Data Preprocessing**

Preprocessing removes errors and handles incomplete information.

Activities include:

- Handling missing patient information.
- Removing duplicate records.
- Correcting invalid values.
- Standardizing patient identifiers.
- Checking inconsistent dates.

**3. Data Transformation**

The selected data is converted into a suitable format for mining.

Examples:

- Calculate patient age.
- Calculate the number of previous admissions.
- Convert diagnosis codes into categories.
- Calculate total medication expenses.
- Create a readmission indicator.

**4. Data Mining**

Data mining algorithms are applied to discover useful patterns.

For MediCore, classification algorithms can predict whether a patient will be readmitted within 30 days.

Suitable algorithms:

- Decision Tree.
- Random Forest.
- Logistic Regression.

**5. Interpretation and Evaluation**

The discovered patterns are evaluated to determine whether they are useful.

For example, the hospital may identify that patients with multiple previous admissions and certain clinical risk factors have a higher readmission rate.

Medical experts should validate these findings before they are used in clinical decision-making.

**2.3 Data Mining System Architecture**

A typical data mining system contains the following components:

1. Database or Data Warehouse Server.

2. Knowledge Base.
3. Data Mining Engine.
4. Pattern Evaluation Module.
5. User Interface.

Figure 2.2: Data Mining System Architecture

**Explanation of Components**

Database / Data Warehouse Server

Stores patient information from different hospital systems.

Knowledge Base

Contains domain knowledge, medical rules, constraints and metadata used to guide the mining process.

Data Mining Engine

Executes mining algorithms such as classification, clustering and association analysis.

Pattern Evaluation Module

Identifies useful and meaningful patterns from the discovered results.

User Interface

Allows doctors, administrators and analysts to interact with the system and view reports.

**2.4 Classification of MediCore Data Mining System**

| Classification | Description | MediCore Application |
|---|---|---|
| Kinds of Data | Relational and warehouse data | Patient, laboratory, pharmacy and billing records |
| Kinds of Knowledge | Classification, association and prediction | Readmission prediction |
| Techniques Used | Machine learning, statistics and database techniques | Decision Tree and Random Forest |
| Applications | Healthcare analytics | Hospital readmission analysis |

**2.5 Data Mining Issues and Solutions**

| Issue | Explanation | Mitigation |
|---|---|---|
| Data Quality | Missing and incorrect patient records | Data cleaning and validation |

| | | |
|---|---|---|
| Methodology | Selecting the correct algorithm | Compare multiple models |
| User Interaction | Medical staff may find results difficult to understand | Provide simple dashboards |
| Efficiency and Scalability | Large patient datasets require processing power | Use indexing and scalable storage |
| Data Diversity | Different systems use different formats | Data integration and standardization |
| Privacy and Security | Patient information is sensitive | Encryption and access control |

**2.6 Difference Between SQL Queries, Statistics and Data Mining**

The hospital director believes that SQL and statistics are sufficient for readmission analysis.

However, these techniques have different purposes.

Database Query Example

SQL can retrieve patients who were readmitted within 30 days.

```
SELECT *
FROM Patient_Admissions
WHERE Readmission_Days <= 30;
```

This retrieves records that satisfy a predefined condition.

Statistical Analysis Example

A statistical summary can calculate the percentage of patients who were readmitted.

$$\text{Readmission Rate} = \frac{\text{Readmitted Patients}}{\text{Total Patients}}\times100$$

Data Mining Example

A classification algorithm can learn from historical patient records and identify combinations of factors associated with future readmission.

It can predict whether a new patient is likely to be readmitted.

Therefore:

- SQL retrieves data.
- Statistics summarizes and measures data.
- Data mining discovers patterns and builds predictive models.

**2.7 Conclusion**

The KDD process helps MediCore convert raw patient data into useful knowledge. Data mining can identify important readmission patterns and support hospital planning, while proper privacy and security measures protect patient information.

**Q3 – Describing and Comparing Data Dispersion (7 Marks)**

**3.1 Introduction**

Data dispersion describes how values are spread around the centre of a dataset.

It helps us understand whether the values are closely grouped or widely distributed.

The given delivery times of 24 orders are:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95.

**3.2 Measures of Central Tendency**

**1. Mean**

Mean is the average of all values.

$$ Mean=\frac{\sum x}{n} $$

Given:

$$ \sum x=1143,\quad n=24 $$

$$ Mean=\frac{1143}{24} $$

Mean = 47.625 minutes

**2. Median**

Median is the middle value of an ordered dataset.

Since there are 24 values, the median is the average of the 12th and 13th values.

$$ Median=\frac{43+45}{2} $$

$$ \boxed{Median=44\text{ minutes}} $$

**3. Mode**

Mode is the value that occurs most frequently.

In the given dataset:

- 35 occurs twice.
- 42 occurs twice.

Therefore:

$$ \boxed{Mode=35,\ 42} $$

The dataset is bimodal.

**4. Midrange**

Midrange is the average of the minimum and maximum values.

$$ Midrange=\frac{Minimum+Maximum}{2} $$

$$ =\frac{28+95}{2} $$

$$ \boxed{Midrange=61.5\text{ minutes}} $$

**Relationship Between Mean and Median**

Mean = 47.625

Median = 44

The mean is greater than the median because the larger delivery time of 95 minutes pulls the mean towards the right.

Therefore, the distribution is positively skewed (right-skewed).

**3.3 Measures of Dispersion**

**1. Range**

Range is the difference between the maximum and minimum values.

$$ Range=Maximum-Minimum $$

$$ =95-28 $$

$$ \boxed{Range=67} $$

**2. Quartiles**

Quartiles divide the dataset into four equal parts.

For this calculation, the median-of-halves method is used.

Lower half:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43

Upper half:

45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95

First Quartile (Q1)

$$ Q1=\frac{36+38}{2} $$

$$ \boxed{Q1=37} $$

Third Quartile (Q3)

$$ Q3=\frac{53+55}{2} $$

$$\boxed{Q3=54}$$

**3. Interquartile Range (IQR)**

IQR measures the spread of the middle 50% of data.

$$IQR=Q3-Q1$$

$$=54-37$$

$$\boxed{IQR=17}$$

**4. Variance**

Variance measures the average squared deviation from the mean.

Treating the 24 observations as the complete population:

$$\sigma^2=\frac{\sum(x-\bar{x})^2}{n}$$

Using the computational formula:

$$\sigma^2=\frac{\sum x^2}{n}-\bar{x}^2$$

Given:

$$\sum x^2=59555$$

$$\sigma^2=\frac{59555}{24}-(47.625)^2$$

$$=2481.4583-2268.1406$$

$$\boxed{\text{Variance}=213.32}$$

**5. Standard Deviation**

Standard deviation is the square root of variance.

$$\sigma=\sqrt{213.32}$$

$$\boxed{\text{Standard Deviation}=14.61}$$

**3.4 Five-Number Summary**

The five-number summary contains:

| Measure | Value |
|---|---|
| Minimum | 28 |
| Q1 | 37 |
| Median | 44 |
| Q3 | 54 |

Maximum 95

**3.5 Boxplot and Outlier Detection**

The 1.5 × IQR rule is used to identify outliers.

Lower fence:

$$ Q1-1.5(IQR) $$

$$ =37-1.5(17) $$

$$ =11.5 $$

Upper fence:

$$ Q3+1.5(IQR) $$

$$ =54+1.5(17) $$

$$ =79.5 $$

Any value below 11.5 or above 79.5 is considered an outlier.

The value 95 is greater than 79.5.

Outlier detected: 95 minutes

Figure 3.1: Boxplot of Delivery Times

**3.6 Recalculation After Removing the Largest Value**

After removing 95, the dataset contains 23 values.

New sum:

$$ 1143-95=1048 $$

New mean:

$$ Mean=\frac{1048}{23} $$

$$ \boxed{45.57} $$

New median:

The 12th value is 43.

$$ \boxed{Median=43} $$

New mode:

$$ \boxed{35,\ 42} $$

New midrange:

$$ \frac{28+72}{2}=50 $$

New range:

$$ 72-28=44 $$

New quartiles:

$$ Q1=36 $$

$$ Q3=52.5 $$

New IQR:

$$ 52.5-36=16.5 $$

New population variance:

$$ \boxed{120.77} $$

New standard deviation:

$$ \boxed{10.99} $$

**Comparison Before and After Removing 95**

| **Measure** | **Before** | **After** |
|---|---|---|
| Mean | 47.625 | 45.565 |
| Median | 44 | 43 |
| Mode | 35, 42 | 35, 42 |
| Midrange | 61.5 | 50 |
| Range | 67 | 44 |
| Q1 | 37 | 36 |
| Q3 | 54 | 52.5 |
| IQR | 17 | 16.5 |
| Variance | 213.32 | 120.77 |
| Standard Deviation | 14.61 | 10.99 |

**Which Measures Changed the Most?**

The range, midrange, variance and standard deviation are strongly affected by removing the largest value.

The mean also changes because it uses every observation.

The median and quartiles change only slightly because they depend on the positions of values rather than their magnitudes.

### 3.7 Classification of Measures

Data mining measures can be classified into three categories.

| Measure | Classification | Explanation |
|---|---|---|
| Mean | Algebraic | Can be calculated using sum and count |
| Median | Holistic | Requires information about ordered values |
| Variance | Algebraic | Uses count, sum and sum of squares |
| Standard Deviation | Algebraic | Calculated from variance |
| Range | Distributive | Uses minimum and maximum |

Distributive measures: Can be calculated from partition summaries and combined directly.

Algebraic measures: Can be calculated using a fixed number of distributive summaries.

Holistic measures: Generally require a larger portion of the original data or a suitable approximate summary.

### 3.8 Tool Verification Using Python

The following Python program verifies the calculations.

```
import numpy as np
from scipy import stats

data = [
   28, 31, 33, 35, 35, 36,
   38, 40, 41, 42, 42, 43,
   45, 46, 48, 50, 52, 53,
   55, 58, 61, 64, 72, 95
]

print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data, keepdims=False))
print("Range:", np.ptp(data))
print("Variance:", np.var(data))
print("Standard Deviation:", np.std(data))

print("After removing 95:")
```

```
new_data = data[:-1]

print("Mean:", np.mean(new_data))
print("Median:", np.median(new_data))
print("Variance:", np.var(new_data))
print("Standard Deviation:", np.std(new_data))
```

The quartile convention used in the manual calculation should be retained when verifying Q1 and Q3.

### 3.9 Recommendation for Typical Delivery Time

The median is recommended as the typical delivery time because it is less affected by unusually long delivery times.

The median delivery time is 44 minutes.

The company should also report the IQR of 17 minutes and the outlier information to understand delivery-time variation.

### 3.10 Conclusion

Measures of central tendency and dispersion help understand delivery performance. The median provides a reliable typical delivery time, while range, variance and standard deviation show how delivery times vary.

## Q4 – Cleaning Noisy and Incomplete Data (6 Marks)

### 4.1 Introduction

Data cleaning is an important preprocessing technique used to identify and correct missing, noisy, inconsistent and invalid data.

A utility company collects hourly smart-meter readings from households. However, the collected data contains several errors.

The objective is to clean the data before performing data mining.

### 4.2 Identification of Data Quality Problems

Given dataset:

| Meter | Household Type | kWh | Voltage | Region |
|---|---|---|---|---|
| M101 | Apartment | 5 | 230 | North |
| M102 | Apartment | Missing | 231 | North |
| M103 | House | 38 | 229 | north |
| M104 | House | -14 | 232 | South |

| | | | | |
|---|---|---|---|---|
| M105 | Missing | 19 | 2300 | South |
| M106 | Apartment | 11 | 230 | S. |

**Defects Identified**

| Meter | Defect | Classification |
|---|---|---|
| M102 | Missing kWh | Missing value |
| M103 | north instead of North | Inconsistency |
| M104 | Negative kWh | Invalid value |
| M105 | Missing household type | Missing value |
| M105 | Voltage 2300 V | Invalid / possible noisy value |
| M106 | S. instead of South | Inconsistency |

The voltage reading of 2300 V is outside the expected range of the other readings and must be investigated.

The 5 kWh reading is not automatically an error because it may represent a valid low-consumption period.

**4.3 Data Cleaning Strategies**

| Problem | Recommended Strategy | Reason |
|---|---|---|
| Missing kWh | Class-wise mean or historical inference | Uses similar meter readings |
| Missing household type | Most probable value | Can be inferred from registered meter details |
| Region inconsistency | Manual standardization | Converts different labels into one format |
| Negative kWh | Ignore or investigate tuple | Invalid consumption value |
| 2300 V | Manual verification / most probable value after verification | May be a measurement or entry error |
| Abnormal readings | Regression or clustering | Identifies unusual patterns |

**Explanation of Cleaning Techniques**

1. Ignore the Tuple

Used when a record contains severe errors that cannot be corrected reliably.

2. Fill Manually

Missing or inconsistent information is corrected using verified source records.

3. Global Constant

A fixed value such as Unknown can be used for missing categorical attributes.

4. Attribute Mean

Missing numerical values are replaced with the average of that attribute.

5. Class-Wise Mean

Missing values are replaced with the mean of records belonging to the same class.

6. Most Probable Value

A missing value is predicted using regression, classification or other inference techniques.

**4.4 Equal-Frequency Binning**

The sorted kWh values of 12 other hourly readings are:

5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38.

Equal-frequency binning divides data into bins containing approximately the same number of observations.

Given:

- Number of values = 12
- Number of bins = 4

Depth:

\[ Depth=\frac{12}{4}=3 \]

Therefore, each bin contains three values.

**Bin Formation**

| Bin | Values |
|---|---|
| Bin 1 | 5, 9, 11 |
| Bin 2 | 14, 18, 19 |
| Bin 3 | 23, 26, 27 |
| Bin 4 | 30, 33, 38 |

**4.5 Smoothing by Bin Means**

Bin means are calculated by finding the average of each bin.

Bin 1

$$ Mean=\frac{5+9+11}{3} $$

$$ =8.33 $$

Bin 2

$$ Mean=\frac{14+18+19}{3} $$

$$ =17 $$

Bin 3

$$ Mean=\frac{23+26+27}{3} $$

$$ =25.33 $$

Bin 4

$$ Mean=\frac{30+33+38}{3} $$

$$ =33.67 $$

**Result of Smoothing by Means**

| Original Bin | Bin Mean | Smoothed Values |
|---|---|---|
| 5, 9, 11 | 8.33 | 8.33, 8.33, 8.33 |
| 14, 18, 19 | 17 | 17, 17, 17 |
| 23, 26, 27 | 25.33 | 25.33, 25.33, 25.33 |
| 30, 33, 38 | 33.67 | 33.67, 33.67, 33.67 |

**4.6 Smoothing by Bin Boundaries**

In this method, each value is replaced with the nearest boundary value of its bin.

The boundaries are the minimum and maximum values of each bin.

| Original Bin | Boundaries | Smoothed Values |
|---|---|---|
| 5, 9, 11 | 5 and 11 | 5, 11, 11 |
| 14, 18, 19 | 14 and 19 | 14, 19, 19 |
| 23, 26, 27 | 23 and 27 | 23, 27, 27 |
| 30, 33, 38 | 30 and 38 | 30, 30, 38 |

For example, in Bin 1:

- 5 is closest to 5.
- 9 is closer to 11.

- 11 is closest to 11.

Therefore, the smoothed bin becomes 5, 11, 11.

**4.7 Use of Regression and Clustering**

Regression

Regression can estimate the expected voltage or electricity consumption based on previous readings and related attributes.

For example, a missing kWh value can be estimated using previous readings from the same household.

Clustering

Clustering groups similar meter readings together.

Readings that are far away from their normal cluster can be identified as possible outliers.

**Why 2300 V Must Be Treated Differently from 5 kWh**

The reading of 2300 V is suspicious because it differs greatly from the other voltage readings, which are approximately 230 V.

It may be caused by an extra zero or measurement error.

However, 5 kWh may be a valid low-consumption value.

Therefore, suspicious values should be checked against domain rules before being replaced or removed.

**4.8 Effect of Global Mean Imputation on Dispersion**

Replacing every missing value with the global mean can reduce the variation in a dataset.

This happens because the mean is usually closer to the centre of the data than extreme values.

As a result:

- Variance may decrease.
- Standard deviation may decrease.
- Differences between records may become smaller.
- Patterns discovered by clustering may become less accurate.
- Relationships between attributes may become distorted.

Global mean imputation should be avoided when the data contains different customer or household groups, because each group may have different average values.

Class-wise mean or predictive imputation may be more suitable in such cases.

**4.9 Conclusion**

Data cleaning improves the accuracy and reliability of mining results. Equal-frequency binning helps smooth noisy data, while regression and clustering can identify abnormal readings. Invalid measurements must be investigated rather than blindly replaced.

**Q5 – Integration and Transformation of Data (7 Marks)**

**5.1 Introduction**

Data integration combines information from multiple sources into a single consistent dataset.

A university combines two systems:

1. Admissions Database.
2. Learning Management System (LMS).

The Admissions database stores student personal information, hostel status and scholarship details.

The LMS stores student names, attendance, marks and scholarship flags.

The purpose is to integrate both systems and prepare a common dataset for analysis.

**Part A – Data Integration**

**5.2 Identification of Data Conflicts**

| **Conflict** | **Admissions Database** | **LMS** | **Resolution** |
|---|---|---|---|
| Entity identification | student_id | sid | Map both to Student_ID |
| Naming conflict | name | full_name | Standardize as Student_Name |
| Date format | Different DOB formats may exist | Different source formats may exist | Convert to YYYY-MM-DD |
| Scholarship conflict | scholarship | scholarship_flag | Convert into a common Yes/No format |
| Unit conflict | Different percentage formats | attendance_pct | Store attendance as percentage |
| Duplicate data | Student name | Full student name | Match using student ID |

**5.3 Integrated Schema**

The combined student database can contain:

| **Attribute** | **Description** |
|---|---|

| | |
|---|---|
| Student_ID | Unique student identifier |
| Student_Name | Student name |
| DOB | Date of birth |
| Hostel_Status | Hostel or Day Scholar |
| Scholarship | Yes or No |
| Attendance_Percentage | Attendance percentage |
| Marks | Academic marks |

This integrated schema provides a single view of student information.

**5.4 Redundant Attributes**

Data integration may create redundant attributes when the same information is available in multiple databases.

For example, student names may be stored in both Admissions and LMS.

Correlation analysis helps identify attributes that are strongly related.

If two attributes provide nearly identical information, one may be unnecessary, depending on its meaning and business importance.

However, high correlation alone does not prove that two attributes are identical.

**Part B – Redundancy Analysis**

**5.5 Pearson Correlation Coefficient**

Given:

Attendance:

60, 70, 75, 80, 85, 90

Marks:

52, 58, 66, 70, 79, 85

Pearson correlation measures the strength and direction of the linear relationship between two variables.

Formula:

$$ r=\frac{\sum(x-\bar{x})(y-\bar{y})} {\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}} $$

The calculated value is:

$$ \boxed{r=0.9851} $$

Very strong positive correlation

**Interpretation**

The correlation coefficient is close to +1.

This indicates a very strong positive linear relationship between attendance and marks.

In general, students with higher attendance tend to have higher marks in this dataset.

However, correlation does not prove that attendance alone causes higher marks.

**5.6 Chi-Square Test**

The chi-square test is used to determine whether two categorical variables are independent.

Given observed values:

| | **Scholarship** | **No Scholarship** | **Total** |
|---|---|---|---|
| Hostel | 120 | 180 | 300 |
| Day Scholar | 80 | 320 | 400 |
| Total | 200 | 500 | 700 |

Formula:

$$E=\frac{Row\ Total\times Column\ Total}{Grand\ Total}$$

**Step 1: Calculate Expected Frequencies**

Hostel and Scholarship:

$$E_{11}=\frac{300\times200}{700}=85.714$$

Hostel and No Scholarship:

$$E_{12}=\frac{300\times500}{700}=214.286$$

Day Scholar and Scholarship:

$$E_{21}=\frac{400\times200}{700}=114.286$$

Day Scholar and No Scholarship:

$$E_{22}=\frac{400\times500}{700}=285.714$$

**Step 2: Expected Frequency Table**

| | **Scholarship** | **No Scholarship** |
|---|---|---|
| Hostel | 85.714 | 214.286 |
| Day Scholar | 114.286 | 285.714 |

**Step 3: Calculate Chi-Square**

Formula:

$$\chi^2=\sum\frac{(O-E)^2}{E}$$

Substituting the observed and expected values:

$$\begin{aligned}\chi^2=&\frac{(120-85.714)^2}{85.714}\\&+\frac{(180-214.286)^2}{214.286}\\&+\frac{(80-114.286)^2}{114.286}\\&+\frac{(320-285.714)^2}{285.714}\end{aligned}$$

$$\boxed{\chi^2=33.60}$$

**Step 4: Hypothesis Testing**

Null hypothesis (H0):

Hostel status and scholarship are independent.

Alternative hypothesis (H1):

Hostel status and scholarship are associated.

Given:

- Calculated chi-square = 33.60
- Critical value = 3.84
- Significance level = 0.05

Since:

$$33.60>3.84$$

We reject the null hypothesis.

Conclusion: Hostel status and scholarship are statistically associated at the 0.05 significance level.

**Part C – Data Transformation**

**5.7 Min-Max Normalization**

Min-max normalization converts data into a specified range, here 0 to 1.

Given annual family incomes:

₹20,000, ₹35,000, ₹50,000, ₹80,000, ₹150,000.

Formula:

$$X'=\frac{X-Min}{Max-Min}$$

Minimum = 20,000

Maximum = 150,000

Range = 130,000

Calculations:

| Income | Calculation | Normalized Value |
|---|---|---|
| 20,000 | (20,000−20,000)/130,000 | 0.0000 |
| 35,000 | 15,000/130,000 | 0.1154 |
| 50,000 | 30,000/130,000 | 0.2308 |
| 80,000 | 60,000/130,000 | 0.4615 |
| 150,000 | 130,000/130,000 | 1.0000 |

**5.8 Z-Score Normalization**

Z-score normalization converts values based on their mean and standard deviation.

$$ Z=\frac{X-\mu}{\sigma} $$

For the given income dataset:

Mean:

$$ \mu=\frac{20000+35000+50000+80000+150000}{5} $$

$$ \mu=67000 $$

Population standard deviation:

$$ \sigma=\sqrt{\frac{\sum(X-\mu)^2}{5}} $$

$$ \sigma\approx46454.28 $$

Normalized values:

| Income | Z-Score |
|---|---|
| 20,000 | -1.0128 |
| 35,000 | -0.6899 |
| 50,000 | -0.3669 |
| 80,000 | 0.2798 |
| 150,000 | 1.7898 |

**5.9 Decimal Scaling Normalization**

Decimal scaling moves the decimal point according to the maximum absolute value.

Formula:

$$ X'=\frac{X}{10^j} $$

Here:

Maximum absolute value = 150,000.

Choose:

$$ j=6 $$

Therefore:

$$ X'=\frac{X}{10^6} $$

| Income | Decimal Scaling Value |
|---|---|
| 20,000 | 0.020 |
| 35,000 | 0.035 |
| 50,000 | 0.050 |
| 80,000 | 0.080 |
| 150,000 | 0.150 |

**5.10 Euclidean Distance Before and After Normalization**

Given two students:

Student A:

$$ (19,35000) $$

Student B:

$$ (21,80000) $$

Euclidean distance formula:

$$ D=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} $$

**Before Normalization**

$$ D=\sqrt{(21-19)^2+(80000-35000)^2} $$

$$ =\sqrt{2^2+45000^2} $$

$$ \boxed{D\approx45000.00004} $$

The income difference dominates the distance because income values are much larger than age values.

**After Min-Max Normalization**

Age range = 17 to 25.

Income range = ₹20,000 to ₹150,000.

For Student A:

$$ Age=\frac{19-17}{25-17}=0.25 $$

$$ Income=\frac{35000-20000}{150000-20000} =0.11538 $$

For Student B:

$$ Age=\frac{21-17}{25-17}=0.50 $$

$$ Income=\frac{80000-20000}{150000-20000} =0.46154 $$

New Euclidean distance:

$$ D=\sqrt{(0.50-0.25)^2+(0.46154-0.11538)^2} $$

$$ D=\sqrt{0.0625+0.11982} $$

$$ \boxed{D\approx0.4270} $$

**Interpretation**

Before normalization, income dominates the distance calculation because it has a much larger numerical scale than age.

After normalization, both attributes contribute more comparably.

This is important for distance-based algorithms such as K-Means and K-Nearest Neighbours.

**5.11 Examples of Data Transformation**

1. Aggregation

Calculate the average marks of all students in each department.

2. Generalization

Convert individual ages into age groups such as:

- Below 20.
- 20–25.
- Above 25.

3. Attribute Construction

Create a new attribute called Academic Performance using attendance and marks.

For example:

\[ Academic\ Score=0.4(Attendance)+0.6(Marks) \]

### 5.12 Conclusion

Data integration combines information from different systems into a consistent dataset. Correlation and chi-square analysis help identify relationships, while normalization makes numerical attributes suitable for distance-based mining algorithms.

## Q6 – Data Reduction and Discretization (7 Marks)

### 6.1 Introduction

Data reduction is the process of reducing the size of a dataset while preserving important information.

In industrial sensor monitoring, a factory collects 200 sensor attributes and approximately 5 million records every month.

Processing all the data directly requires more storage and computational resources.

Data reduction and discretization help improve processing speed while retaining important information needed to detect machine failures.

## Part A – Choosing Data Reduction Strategies

### 6.2 Data Reduction Techniques

1. Data Cube Aggregation

Data cube aggregation summarizes detailed data into higher-level information.

Example:

Instead of analysing every second of machine temperature readings, the system can calculate average temperature for each hour or day.

This reduces the number of records that need to be analysed.

2. Attribute Subset Selection

Attribute subset selection identifies the most useful attributes and removes irrelevant or redundant ones.

Methods:

- Forward Selection: Starts with no attributes and adds useful attributes.
- Backward Elimination: Starts with all attributes and removes less useful attributes.
- Decision Tree Induction: Identifies important attributes based on their contribution to classification.

Example: Select temperature, vibration and pressure if they provide useful information for predicting machine failures.

3. Dimensionality Reduction – PCA

Principal Component Analysis (PCA) reduces the number of variables by transforming correlated attributes into a smaller set of principal components.

Example:

200 sensor attributes can be converted into a smaller number of principal components while retaining most of the variation.

4. Numerosity Reduction

Numerosity reduction represents a large dataset using a smaller representation.

Methods include:

- Sampling.
- Histograms.
- Clustering.

Example: A representative sample of sensor readings can be used for preliminary analysis.

5. Data Compression

Compression reduces the storage space required for data.

It can be:

- Lossless compression: Preserves all original information.
- Lossy compression: Removes some information to reduce storage.

For failure detection, lossless compression or carefully validated lossy compression should be preferred.

**6.3 Comparison of Sampling Methods**

| Sampling Method | Description | Advantage | Disadvantage |
|---|---|---|---|
| Random Sampling Without Replacement | Selected records cannot be selected again | No duplicate samples | Rare failures may be missed |
| Random Sampling With Replacement | A record may be selected more than once | Simple sampling process | Duplicate records may occur |
| Cluster Sampling | Selects complete groups of records | Useful for grouped data | Selected groups may not represent the entire population |

| | | | |
|---|---|---|---|
| Stratified Sampling | Divides data into groups and samples from each group | Preserves group representation | Requires prior class information |

**6.4 Best Sampling Method for Rare Failures**

Given:

- Failure records = 0.5%.
- Normal records = 99.5%.

The most suitable method is Stratified Sampling.

In this method, data is divided into two groups:

1. Failure records.
2. Normal records.

Samples are selected from both groups.

Since failures represent only 0.5% of the data, ordinary random sampling may select very few failure records.

Stratified sampling ensures that sufficient failure examples are available for model training.

The final test dataset should retain the real failure proportion to evaluate actual performance.

**6.5 Selection of Principal Components**

PCA should retain enough principal components to explain at least 95% of the cumulative variance as an initial criterion.

For example, if 35 principal components explain 95% of the original variation, those components may be selected instead of using all 200 attributes.

However, in machine failure detection, rare failure information must not be lost.

Therefore, the selected components should also be tested using failure-detection metrics such as recall and precision.

**Part B – Discretization by Calculation**

**6.6 Given Temperature Readings**

3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44.

Total values:

$$ n=20 $$

Number of intervals:

$$ k=4 $$

### 6.7 Equal-Width Binning

Equal-width discretization divides the complete data range into intervals of equal size.

Formula:

$$ Width=\frac{Maximum-Minimum}{Number\ of\ bins} $$

Given:

Maximum = 44

Minimum = 3

Number of bins = 4

$$ Width=\frac{44-3}{4} $$

$$ \boxed{Width=10.25} $$

**Interval Formation**

| Bin | Interval | Values | Count |
|---|---|---|---|
| Bin 1 | 3–13.25 | 3, 7, 7, 9, 12 | 5 |
| Bin 2 | 13.25–23.5 | 14, 14, 15, 18, 21, 21, 22 | 7 |
| Bin 3 | 23.5–33.75 | 25, 25, 27, 31 | 4 |
| Bin 4 | 33.75–44 | 36, 36, 40, 44 | 4 |

The number of values in each interval is not equal because equal-width binning uses the same interval size rather than the same number of observations.

### 6.8 Equal-Frequency Binning

Equal-frequency binning divides the dataset into intervals containing approximately the same number of values.

Given:

$$ n=20,\quad k=4 $$

$$ Depth=\frac{20}{4}=5 $$

Therefore, each bin contains five values.

| Bin | Values | Count |
|---|---|---|
| Bin 1 | 3, 7, 7, 9, 12 | 5 |
| Bin 2 | 14, 14, 15, 18, 21 | 5 |

Bin 3 21, 22, 25, 25, 27 5

Bin 4 31, 36, 36, 40, 44 5

**6.9 Entropy-Based Discretization**

Entropy is used to measure the impurity or uncertainty of class labels.

The given dataset is:

**Value 10 12 15 18 22 25 28 31 35 40**

Class N N N N Y Y N Y Y Y

There are:

- Negative (N) = 5
- Positive (Y) = 5

Entropy formula:

$$ H(S)=-p_N\log_2p_N-p_Y\log_2p_Y $$

Initial entropy:

$$ H(S)=-\frac55\log_2\frac55 $$

Correction: using the total of 10 records, the two class proportions are each 0.5.

$$ H(S)=-0.5\log_2(0.5)-0.5\log_2(0.5) $$

$$ \boxed{H(S)=1} $$

**Candidate Split Points**

The candidate split points are the midpoints between adjacent values.

| Adjacent Values | Split Point |
|---|---|
| 10 and 12 | 11 |
| 12 and 15 | 13.5 |
| 15 and 18 | 16.5 |
| 18 and 22 | 20 |
| 22 and 25 | 23.5 |
| 25 and 28 | 26.5 |
| 28 and 31 | 29.5 |

| | |
|---|---|
| 31 and 35 | 33 |
| 35 and 40 | 37.5 |

Information Gain:

$$ IG=H(S)-H(Split) $$

Where:

$$ H(Split)= \frac{|S_1|}{|S|}H(S_1) + \frac{|S_2|}{|S|}H(S_2) $$

**Information Gain Calculation**

| Split Point | Information Gain |
|---|---|
| 11 | 0.1080 |
| 13.5 | 0.2365 |
| 16.5 | 0.3958 |
| 20 | 0.6100 |
| 23.5 | 0.2781 |
| 26.5 | 0.1245 |
| 29.5 | 0.3958 |
| 33 | 0.2365 |
| 37.5 | 0.1080 |

**Best Split Point**

The highest information gain is obtained at split point 20.

$$ \boxed{Best\ Split=20} $$

$$ \boxed{Information\ Gain=0.6100} $$

This split separates the first four negative records from the remaining records, producing the greatest reduction in class uncertainty.

**6.10 Unsupervised vs Supervised Discretization**

| Feature | Unsupervised Discretization | Supervised Discretization |
|---|---|---|
| Class labels | Not required | Required |
| Purpose | Divide values into intervals | Find intervals useful for classification |

| | | |
|---|---|---|
| Methods | Equal-width and equal-frequency | Entropy-based discretization |
| Knowledge used | Distribution of data | Distribution and class labels |
| Example | Temperature grouping | Machine failure prediction |

**Which Is Better for Classification?**

Supervised discretization is more suitable for classification because it considers the class labels while creating intervals.

For example, entropy-based discretization identifies temperature ranges that best distinguish normal machine operation from failure conditions.

**6.11 Conclusion**

Data reduction reduces the size of large sensor datasets while preserving useful information. Sampling, PCA, aggregation and compression improve processing efficiency. Discretization converts continuous readings into intervals, while entropy-based methods support accurate classification.

**Q7 – Diagnose, Redesign and Validate (Day-4 Design Challenge – 13 Marks)**

**7.1 Introduction**

A retail bank developed a machine learning model to predict loan defaults.

However, the model performed poorly after deployment even though it reported 97% accuracy during testing.

The main reason is that the original preprocessing pipeline contains several problems.

The objective is to identify these problems, redesign the preprocessing process and develop a more reliable model.

**A. Diagnose the Computational Bottleneck (2 Marks)**

**7.2 Problems in the Original Pipeline**

The original pipeline contains five major preprocessing decisions.

| Original Step | Problem | Principle Violated | Effect |
|---|---|---|---|
| Missing income replaced with global mean | Ignores different income groups | Appropriate imputation | Reduces variation and may distort relationships |
| Occupation with 18% missing dropped | Removes potentially useful information | Feature preservation | Loses important customer information |

| | | | |
|---|---|---|---|
| Income divided into five equal-width intervals | Income is highly right-skewed | Appropriate discretization | Creates unbalanced intervals |
| Normalization before train-test split | Uses information from test data | Data leakage prevention | Produces overly optimistic evaluation |
| Random 10% training sample | Only 3% of loans are defaults | Class imbalance handling | May contain too few defaulters |

**Explanation**

1. Global Mean Imputation

Replacing all missing income values with one global mean reduces variation and may not represent different customer groups.

2. Dropping Occupation

Occupation may be useful for predicting loan defaults. Removing it completely can reduce predictive information.

3. Equal-Width Discretization

Since income is highly right-skewed, equal-width intervals may create bins containing very different numbers of records.

4. Data Leakage

Normalization was performed before splitting the data.

This means information from the test dataset was used during preprocessing.

As a result, the model evaluation may become unreliable.

5. Class Imbalance

Only 3% of loans are defaults.

If 10% of all records are selected randomly for training, the sample may contain too few default records.

This makes the model biased towards predicting non-default cases.

**Why 97% Accuracy Is Misleading**

Suppose there are 10,000 loan records.

Default records:

\[ 10,000\times0.03=300 \]

Non-default records:

\[ 10,000-300=9700 \]

A model that predicts every loan as non-default achieves:

$$Accuracy=\frac{9700}{10000}\times100$$

$$\boxed{Accuracy=97\%}$$

However, it fails to identify every actual defaulter.

Therefore, accuracy alone is not a reliable evaluation metric for imbalanced datasets.

**B. Define the Design Objective (2 Marks)**

**7.3 Objectives of the Redesigned Pipeline**

The redesigned pipeline should improve the identification of defaulting customers.

The main objectives are:

1. Increase recall for the default class.
2. Improve precision to reduce unnecessary false alarms.
3. Reduce data leakage.
4. Preserve useful information from missing occupation values.
5. Maintain the original test-data distribution.
6. Improve the reliability of model predictions.

**Performance Metrics**

1. Accuracy

Measures the percentage of correctly classified records.

$$Accuracy=\frac{TP+TN}{TP+TN+FP+FN}$$

2. Precision

Measures how many predicted defaults are actually defaults.

$$Precision=\frac{TP}{TP+FP}$$

3. Recall

Measures how many actual defaults are correctly identified.

$$Recall=\frac{TP}{TP+FN}$$

4. F1-Score

Balances precision and recall.

$$F1=2\times\frac{Precision\times Recall}{Precision+Recall}$$

**Recommended Design Targets**

The following are proposed objectives, not measured results:

| Metric | Proposed Target |
|---|---|
| Default Recall | At least 80% |
| Precision | At least 50% |
| F1-Score | Improve over baseline |
| Data Leakage | Zero |
| Test Dataset | Unchanged and stratified |
| Processing | Reproducible pipeline |

The bank should prioritize recall because missing a defaulter can cause financial losses. However, precision must also be considered because too many false alarms can lead to unnecessary loan rejections.

**C. Design the Revised Preprocessing Strategy (4 Marks)**

**7.4 Proposed Revised Pipeline**

The redesigned pipeline follows an ordered sequence.

Figure 7.1: Redesigned Loan Default Prediction Pipeline

**Step 1 – Data Collection and Inspection**

Collect loan information such as:

- Customer age.
- Income.
- Occupation.
- Loan amount.
- Credit history.
- Employment status.
- Repayment history.
- Loan default status.

Identify missing values, duplicate records and invalid values.

**Step 2 – Train-Test Split**

Split the dataset before performing normalization or other learned preprocessing.

Use stratified splitting to maintain the class distribution.

Example:

- Training data = 80%.
- Testing data = 20%.

Stratification ensures that both sets contain approximately the same proportion of default and non-default records.

**Step 3 – Missing Value Handling**

For missing income:

Use median imputation or class-independent group-based imputation where justified.

Median is preferred over mean because income is highly right-skewed and median is less affected by extreme values.

For occupation:

- Retain the attribute.
- Replace missing values with Unknown.
- Optionally use a predictive imputation method if sufficient reliable information exists.

**Step 4 – Data Transformation**

Categorical attributes should be converted into numerical representations.

Examples:

- Occupation.
- Employment status.
- Credit history category.

One-hot encoding can be used for categorical variables.

Numerical attributes should be normalized or standardized when required by the selected algorithm.

**Step 5 – Income Discretization**

Instead of equal-width discretization, use equal-frequency binning or an appropriate supervised discretization method.

Equal-frequency binning creates intervals containing approximately equal numbers of records.

This is more suitable for highly skewed income data.

**Step 6 – Attribute Selection**

Select useful features using:

- Correlation analysis.
- Decision Tree feature importance.
- Information Gain.

Irrelevant and redundant attributes can be removed after validation.

**Step 7 – Class Imbalance Handling**

Use stratified sampling and class-weighted learning.

If additional resampling is necessary, techniques such as SMOTE can be applied only to the training data.

The test dataset must retain the original 3% default distribution.

**Step 8 – Model Training**

Suitable algorithms include:

1. Random Forest.
2. Logistic Regression.
3. Decision Tree.

Random Forest can identify complex relationships, while Logistic Regression provides a simpler baseline.

**Step 9 – Model Evaluation**

Evaluate the model using:

- Accuracy.
- Precision.
- Recall.
- F1-Score.
- Confusion Matrix.
- ROC-AUC and PR-AUC.

For an imbalanced default-prediction problem, PR-AUC and recall are particularly useful.

**Comparison with Alternatives**

| Problem | Selected Technique | Alternative | Reason |
|---|---|---|---|
| Missing income | Median imputation | Mean imputation | More robust to skewness |

| | | | |
|---|---|---|---|
| Missing occupation | Unknown category | Drop attribute | Preserves useful information |
| Income discretization | Equal-frequency | Equal-width | Better handles skewed data |
| Data leakage | Split before scaling | Scale entire dataset | Prevents test information leakage |
| Imbalanced data | Class weighting | Ordinary random sampling | Gives importance to minority class |
| Feature selection | Tree-based importance | Keep all features | Reduces irrelevant attributes |

**Training-Only Processing**

The following operations must be fitted using training data only:

- Missing-value imputation.
- Scaling and normalization.
- Feature selection.
- Discretization when learned from data.
- Oversampling and SMOTE.
- Model training.

The fitted transformations are then applied to the test data without refitting.

**D. Predict Before Implementing (1 Mark)**

**7.5 Expected Changes After Redesign**

Before running the redesigned model, the following changes are expected.

| **Parameter** | **Expected Change** | **Reason** |
|---|---|---|
| Income Dispersion | Better preserved | Median imputation and suitable transformations |
| Training Class Distribution | More balanced if training-only resampling is applied | Improves minority-class learning |
| Accuracy | May decrease | Model no longer simply favours the majority class |

| | | |
|---|---|---|
| Default Recall | Expected to increase | Class weighting and better training representation |
| Precision | May decrease or increase depending on threshold | More default cases may be detected, with possible false positives |

The main expectation is that the redesigned model will identify more actual defaulters, even if overall accuracy decreases slightly.

**E. Implement and Validate the Redesign (2 Marks)**

**7.6 Python Implementation**

The following Python program demonstrates a revised preprocessing and prediction pipeline using Scikit-learn.

It assumes a CSV dataset named loan_data.csv with a target column named default and example numerical and categorical columns.

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix
)

# Load dataset
data = pd.read_csv("loan_data.csv")

# Separate input and output
X = data.drop("default", axis=1)
y = data["default"]

# Identify column types
num_cols = X.select_dtypes(
    include=["int64", "float64"]
).columns
```

```
cat_cols = X.select_dtypes(
    include=["object", "category"]
).columns

# Numerical preprocessing
num_pipe = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

# Categorical preprocessing
cat_pipe = Pipeline([
    ("imputer", SimpleImputer(
        strategy="constant",
        fill_value="Unknown"
    )),
    ("encoder", OneHotEncoder(
        handle_unknown="ignore"
    ))
])

# Combine preprocessing
preprocessor = ColumnTransformer([
    ("num", num_pipe, num_cols),
    ("cat", cat_pipe, cat_cols)
])

# Create model
model = Pipeline([
    ("preprocessing", preprocessor),
    ("classifier", RandomForestClassifier(
        n_estimators=100,
        class_weight="balanced",
        random_state=42
    ))
])

# Stratified train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    stratify=y,
```

```
    random_state=42
)

# Train model
model.fit(X_train, y_train)

# Predict
predictions = model.predict(X_test)

# Evaluate
print("Accuracy:",
      accuracy_score(y_test, predictions))

print("Precision:",
      precision_score(y_test, predictions,
                      zero_division=0))

print("Recall:",
      recall_score(y_test, predictions,
                   zero_division=0))

print("F1 Score:",
      f1_score(y_test, predictions,
               zero_division=0))

print("Confusion Matrix:")
print(confusion_matrix(y_test, predictions))
```

**Explanation of the Program**

1. The dataset is loaded using Pandas.
2. Input attributes and the target variable are separated.
3. Numerical and categorical attributes are identified.
4. Missing numerical values are filled using the median.
5. Missing categorical values are filled with Unknown.
6. Numerical attributes are standardized.
7. Categorical attributes are encoded.
8. The dataset is divided using stratified sampling.
9. A Random Forest model is trained with class weighting.

10. Predictions are evaluated using different metrics.

The preprocessing operations are fitted only on the training dataset through the Scikit-learn pipeline.

This is a redesigned-pipeline implementation template. The exact CSV column names and target encoding must match the selected dataset. It demonstrates the core preprocessing approach, but does not implement every optional technique discussed above, such as SMOTE or supervised income discretization.

### 7.7 Original vs Redesigned Pipeline Comparison

The assignment requires both pipelines to be implemented and evaluated on the same test data.

Use the following result table:

| Evaluation Metric | Original Pipeline | Redesigned Pipeline |
|---|---|---|
| Accuracy | 97% (given) | Actual result |
| Precision | Measure | Actual result |
| Recall | Measure | Actual result |
| F1-Score | Measure | Actual result |
| Confusion Matrix | Generate | Generate |

The original 97% accuracy is given in the question. Its other metrics are not provided and must be calculated from the original model's predictions.

Do not invent the redesigned model's performance values.

## F. Defend the Final Design (2 Marks)

### 7.8 Comparison and Justification

The redesigned pipeline is more reliable because it addresses the main weaknesses of the original preprocessing process.

| Feature | Original Pipeline | Redesigned Pipeline |
|---|---|---|
| Income Imputation | Global mean | Median |
| Occupation | Dropped | Retained with Unknown category |
| Discretization | Equal-width | Equal-frequency or supervised |
| Normalization | Before splitting | After splitting |

| | | |
|---|---|---|
| Sampling | Random 10% | Stratified train-test split |
| Class Imbalance | Ignored | Class-weighted learning |
| Evaluation | Accuracy | Multiple evaluation metrics |

**Advantages of the Redesigned Pipeline**

1. Reduces the possibility of data leakage.
2. Preserves important customer information.
3. Handles missing income values more appropriately.
4. Improves the representation of minority-class information.
5. Provides a more reliable evaluation.
6. Focuses on identifying actual defaulters.

**Remaining Risk**

One important remaining risk is that customer financial behaviour may change over time.

A model trained using historical loan information may become less accurate when economic conditions, interest rates or customer repayment patterns change.

Therefore, the bank should regularly monitor model performance and retrain the model when necessary.

**7.9 Final Conclusion**

The original loan-default prediction pipeline produced misleading results because of incorrect preprocessing, data leakage and class imbalance.

Although it reported 97% accuracy, it failed to identify most actual defaulters.

The redesigned pipeline uses median imputation, suitable discretization, stratified sampling, training-only normalization and class-weighted Random Forest classification.

It is expected to improve the identification of defaulting customers and provide a more reliable evaluation.

Therefore, the redesigned pipeline is more appropriate for loan-default prediction because it focuses on data quality, fair evaluation and the detection of minority-class records.

**OVERALL ASSIGNMENT CONCLUSION**

Data mining plays an important role in discovering useful information and patterns from large datasets. This assignment explains different data mining functionalities, preprocessing methods and techniques used to improve data quality.

The TeleNova case study demonstrates how association, classification, clustering, outlier analysis and evolution analysis can solve different business problems.

The MediCore case study explains how the KDD process converts raw hospital information into meaningful knowledge. It also highlights the importance of data mining architecture, privacy and security.

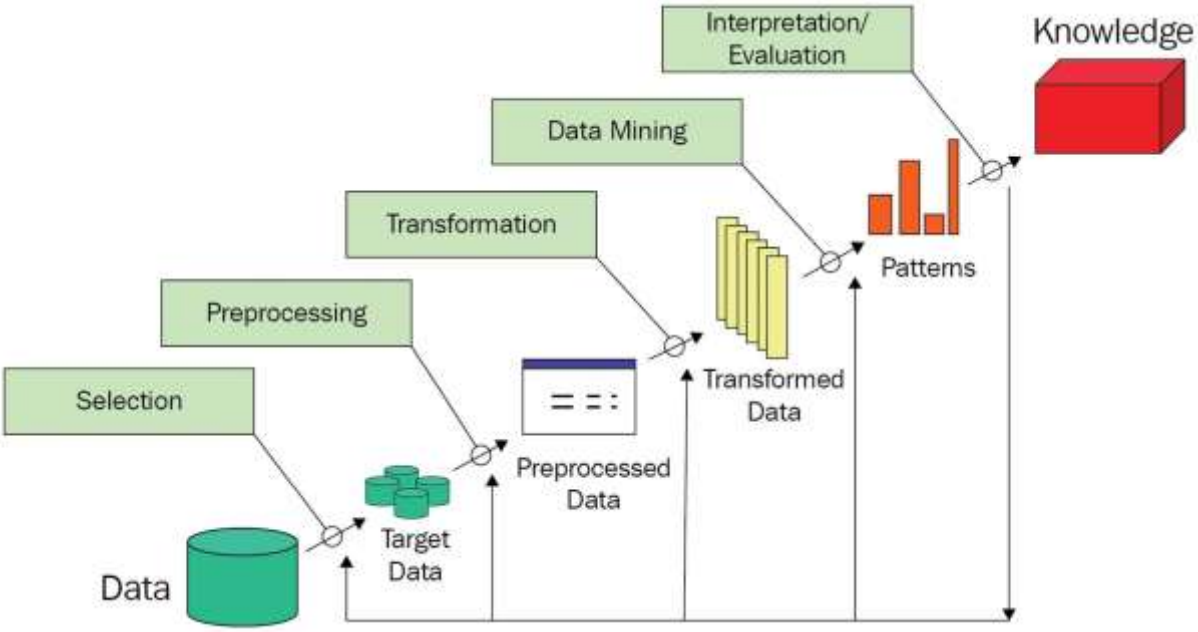
The delivery-time analysis demonstrates how mean, median, mode, variance, standard deviation and quartiles are used to understand data dispersion. Data cleaning techniques help identify and correct missing, inconsistent, noisy and invalid values.

Data integration and transformation combine information from different sources and prepare it for mining. Normalization methods such as min-max, z-score and decimal scaling improve the quality of distance-based algorithms.

Data reduction and discretization reduce processing complexity while preserving important information. Finally, the loan-default prediction case study demonstrates how a properly designed preprocessing pipeline improves model reliability and evaluation.

Thus, data mining and preprocessing techniques help organizations make accurate predictions, discover useful patterns and support better decision-making.

Figure 2.1: KDD Process



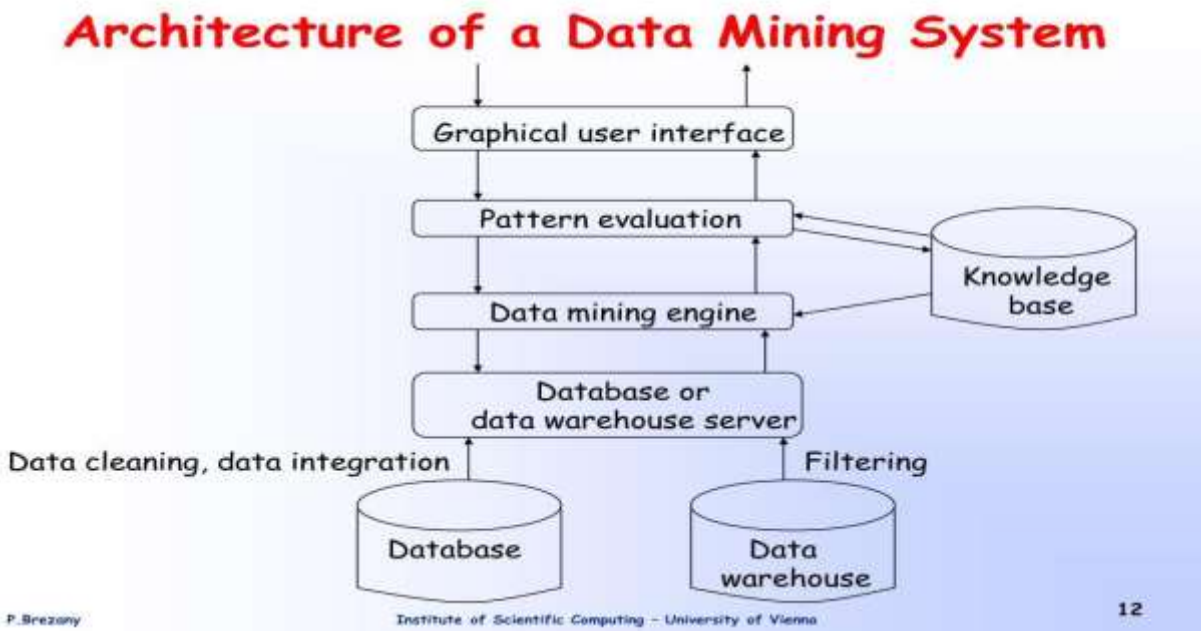
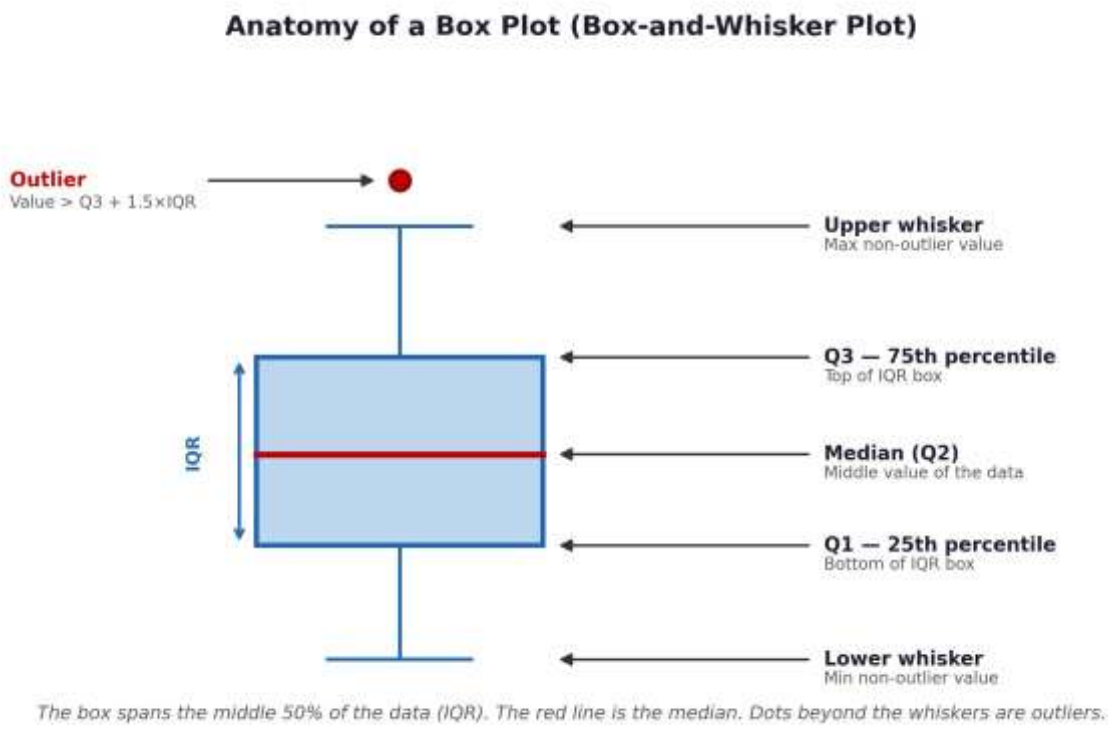




Figure 2.2: Data Mining System Architecture

Figure 3.1: Boxplot of Delivery Times



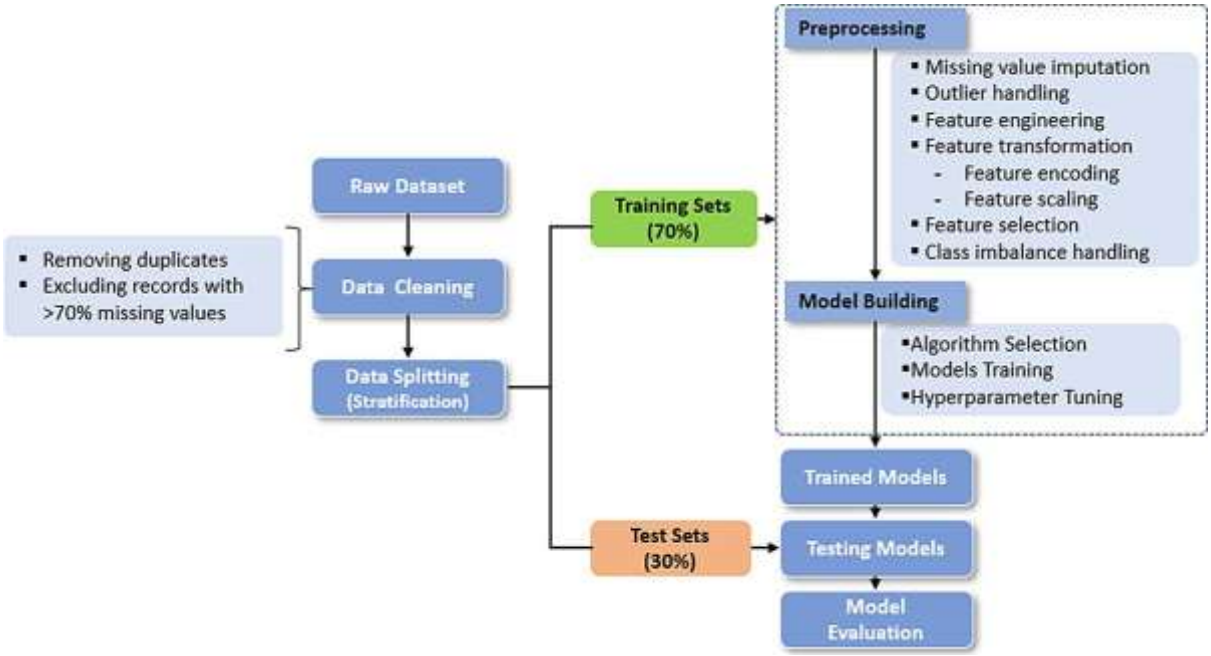




Figure 7.1: Redesigned Loan Default Prediction Pipeline